

## **BigDecimal v1.4 Add-in**

Functional Understanding Document

Version 1.2

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# 1 Introduction

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Welcome to BigDecimal, a high-performance, arbitrary-precision decimal library designed to break the 15-digit precision barrier of standard Excel calculations. Heavily inspired by the robust BigDecimal implementations found in enterprise languages like Java, this add-in brings industrial-grade numerical accuracy directly to your spreadsheets.

Whether you are performing complex financial reconciliations or deep scientific research, BigDecimal ensures that your results are never compromised by floating-point errors or rounding limitations.

## 1.1 Why BigDecimal?

Standard Excel uses the IEEE 754 floating-point standard, which is excellent for speed but can introduce "silent" errors in very large or very small numbers. BigDecimal for Excel replaces this with an exact-value representation, allowing you to define the exact scale and precision required for your specific use case.

# 2 The BigCalculator: Desktop-Grade Precision in Excel

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Serving as the flagship showcase for the **BigDecimal** add-in, the **BigCalculator** (Figure 1) is a fully functional, real-time application designed to demonstrate the sheer speed and raw numerical depth of the underlying mathematical kernel. Boasting a large, highly legible display, it provides a user-friendly interface for executing hyper-precise calculations that easily transcend the limitations of traditional 64-bit hardware.

The calculator exposes a comprehensive suite of advanced mathematical tools directly on its control panel: standard operations (+, -, ×, ÷), algebraic controls (Absolute Value, Sign Inversion, Truncation, Fraction Splitting, Modulo, and Strict Comparisons), alongside rapid transcendental engines like powers ( $x^y$ ), Natural and Base-10 Logarithms ( $\ln$ ,  $\log$ ), Exponentials ( $\exp$ ), Square Roots ( $\sqrt{x}$ ), Factorials ( $n!$ ), and instant, arbitrary-scale generations of mathematical constants like  $\pi$  and  $e$ .

To give developers absolute control over the execution layout, the application features an integrated Precision **Scale** button and an editable configuration box. Users can dynamically input their exact target decimal scale right on the interface, instantly recalibrating the calculator to process complex formulas across dozens or hundreds of digits with absolute, character-for-character fidelity. As the official example workbook, the **BigCalculator** stands as concrete proof that Excel can be transformed into a desktop-grade, infinite-precision computing station.

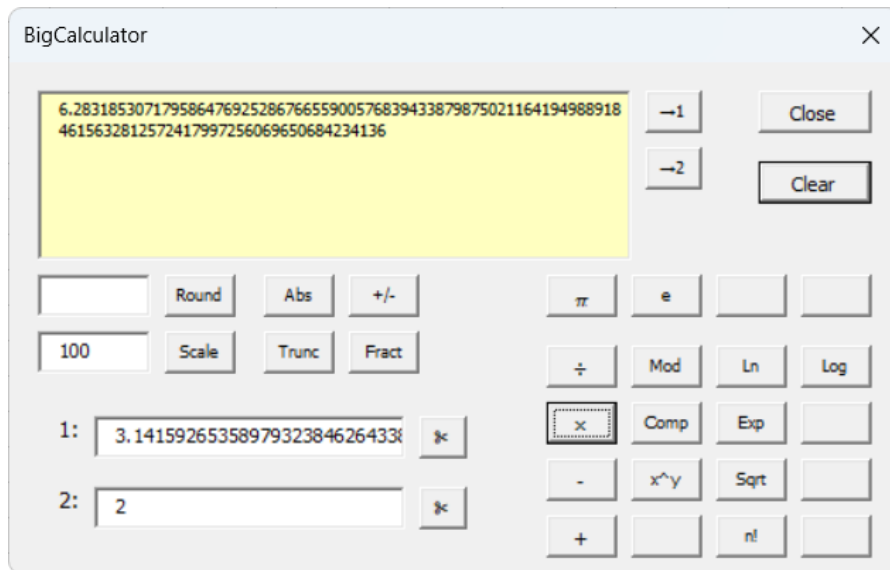


Figure 1: The BigCalculator

### 3 Installation of BigDecimal v1.4.xlam

Follow the steps below to install the **BigDecimal** add-in and integrate it with Microsoft Excel. These steps ensure the library is correctly registered and available for use within the VBA Editor.

#### 3.1.1 Downloading the BigDecimal Add-in

To install the Add-in, download the **BigDecimal v1.4.xlam** file from the [GitHub](#) repository and move it to the local Microsoft Add-ins directory:

C:\Users\[YourUsername]\AppData\Roaming\Microsoft\AddIns

*Note: Replace [YourUsername] with your active Windows login name.*

#### Quick Access Shortcut:

Alternatively, you can access this folder directly by pasting the following environment variable into the File Explorer address bar:

%appdata%\Microsoft\AddIns

Once the folder is open, drag and drop the BigDecimal v1.4.xlam file from your **Downloads** folder into the **Addins** folder as in Figure 3-1.

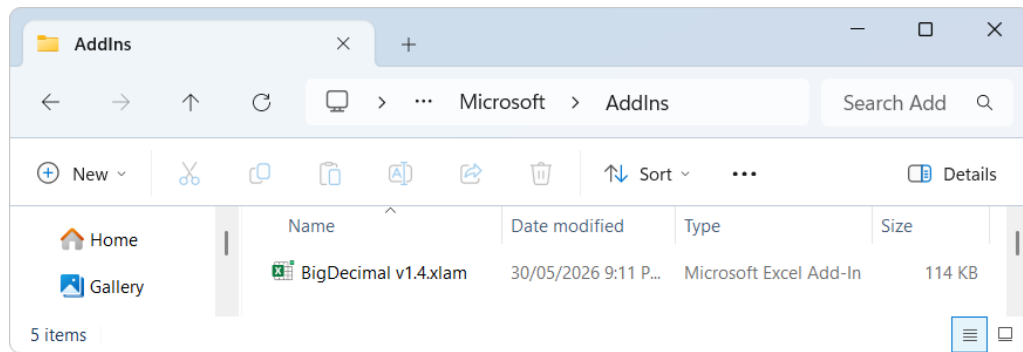


Figure 3-1: Excel Add-ins Folder

### 3.1.2 Important: Unblocking the Add-in

Because the file was downloaded from an external source (GitHub), Windows will apply a 'Mark of the Web' security attribute, which prevents the Add-in from executing macros. Even if the file is placed in the correct directory, it will not function until it is manually unblocked.

#### To unblock the Add-in:

1. Navigate to the %appdata%\Microsoft\AddIns folder
2. **Right-click** the BigDecimal v1.4.xlam file.
3. Select **Properties**.
4. At the bottom of the **General** tab, look for **Security**.
5. Check the **Unblock** box and click **Apply**.

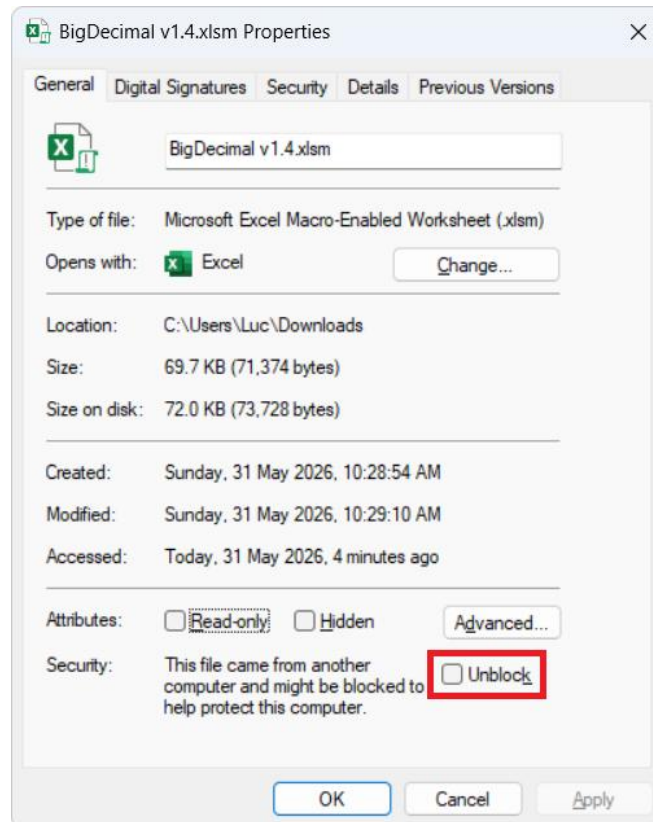


Figure 3-2: Unblock Downloaded BigDecimal Add-in

## 4 Working with BigDecimal

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Before the **BigDecimal** class can be utilized in your VBA project, it must be properly initialized within the VBA environment. This involves linking the Add-in to your specific workbook and ensuring the VBA compiler can access the library's properties and methods.

### 4.1 Referencing BigDecimal in VBA

Once the Add-in is installed and unblocked, you must establish a reference to the library within your specific VBA Project. This allows the VBA compiler to recognize the **BigDecimal** class and provide IntelliSense support while coding.

#### Follow these steps to enable the reference:

1. Open Excel.
2. Go to the **Developer** tab (If you don't see it, right-click any ribbon tab > Customize the Ribbon > Check "Developer").
3. Click **View Code**.
4. Go to the **Tools** menu > **References..** and locate BigDecimal\_v1\_4 and select it as in Figure 4-1. A tick mark will appear next to BigDecimal\_v1\_4.
5. Click **OK**.

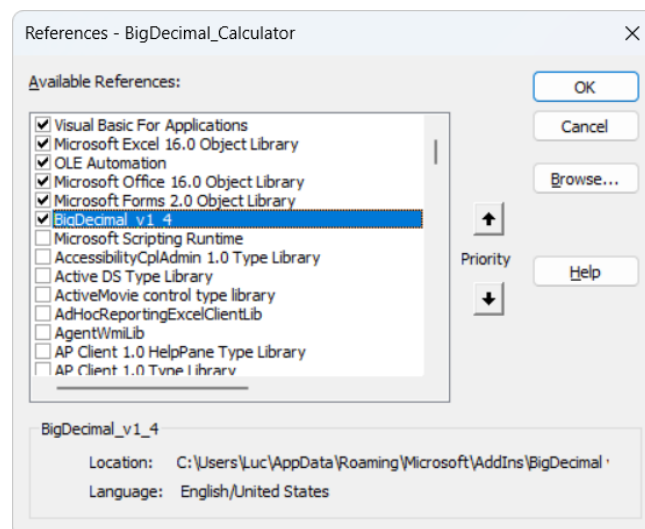


Figure 4-1: Adding BigDecimal\_v1\_4 as a Reference

At this stage the add-in is now available to use in your VBA Project.

## 5 BigDecimal “Quick Start” Guide

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BigDecimal is designed to work seamlessly within the Excel environment, its implementation respects the standard VBA architectural patterns used by Microsoft Office. To ensure maximum compatibility and performance across different workbooks, the library utilizes a

**controlled instantiation model.** This approach prevents common reference errors and ensures that every `BigDecimal` object is correctly initialized with the high-precision engine.

### 5.1 Instantiating the `BigDecimal` Class

In standard object-oriented programming, you would typically instantiate a class using the `New` keyword. However, due to the **PublicNotCreatable** nature of classes within an Excel Add-in environment, the `BigDecimal` class cannot be created directly by external projects.

To bypass this limitation, the Add-in provides a global factory function: `New_BigDecimal()`. This function serves as the entry point for object creation, allowing you to initialize a new `BigDecimal` instance. It optionally accepts a string parameter to set the initial value of the number, ensuring precision is maintained from the moment of instantiation.

```
Public Function New_BigDecimal(Optional StrValue As String = "") As  
    BigDecimal
```

### 5.2 Casting standard types

To simplify workflows, the Add-in includes the `CBgDec()` casting function. This function mirrors native VBA conversion functions (like `Cdbl` or `CDec`), accepting standard data types and returning a fully initialized `BigDecimal` object.

```
Public Function CBgDec(Value As Variant) As BigDecimal
```

#### Supported Conversions

The `CBgDec()` function safely handles the following native types by converting them to their string representations internally before initializing the `BigDecimal` object:

- **Numeric Types:** Integer, Long, LongLong, Single, Double.
- **Strings:** Standard text representations of numbers (e.g., "123.4567890123456789").

### 5.3 Implementation Example: Basic Addition

The following example, adapted from the **BigCalculator** source code, demonstrates how to instantiate two `BigDecimal` objects from user input, perform basic validation, and output the result of an addition operation.



```

Private Sub Add_Click()
    Dim Num1 As BigDecimal
    Dim Num2 As BigDecimal

    With BigDecimalForm
        ' Instantiate objects using the New_BigDecimal factory function
        Set Num1 = New_BigDecimal(.Num1.Value)
        Set Num2 = New_BigDecimal(.Num2.Value)

        ' Validate inputs and check for successful instantiation
        If .Num1.Value = "" Then
            ErrBox "Number 1 is empty!", "Input Error"
            Exit Sub
        ElseIf .Num2.Value = "" Then
            ErrBox "Number 2 is empty!", "Input Error"
            Exit Sub
        ElseIf Num1 Is Nothing Then
            ErrBox "Number 1 is invalid!", "Validation Error"
            Exit Sub
        ElseIf Num2 Is Nothing Then
            ErrBox "Number 2 is invalid!", "Validation Error"
            Exit Sub
        End If

        ' Perform addition and return the result as a string
        .Answer.Value = Num1.Plus(Num2).ValueEx(vbString)
    End With
End Sub

```

#### 5.4 Language Limitations: Operator Overloading

A notable constraint of the VBA environment is the lack of **Operator Overloading**. In many modern languages, you can redefine standard operators (such as +, -, \*, or /) to work directly with custom classes. Ideally, one would use a syntax like `Set Result = Num1 + Num2`; however, this is not supported in VBA.

Consequently, there is no direct workaround for this language limitation. All arithmetic operations must be performed by calling the explicit methods of the `BigDecimal` class. While this requires a slightly more verbose coding style, the method names are intuitive and follow standard mathematical naming conventions.

## 6 Functional Specifications

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This section provides a detailed technical breakdown of the public classes, methods, and properties that comprise the **BigDecimal** framework. These components form the core engine of the library, enabling the programmatic initialization of high-precision variables, the execution of complex mathematical operations, and the fine-tuning of calculation parameters such as scale and rounding. By leveraging these members, developers can extend the numerical capabilities of their Excel applications far beyond the standard limits of floating-point arithmetic.

## 6.1 BigDecimal Methods

The following table outlines the comprehensive suite of arithmetic and utility methods supported by the **BigDecimal** class, designed to provide consistent results for high-precision computational tasks.

| Methods | VBA Syntax (BigDecimal) | Notes                                                |
|---------|-------------------------|------------------------------------------------------|
| Plus    | Num1.Plus(Num2)         | Returns the sum of Num1 and Num2.                    |
| Minus   | Num1.Minus(Num2)        | Returns the difference (Num1 - Num2).                |
| Times   | Num1.Times(Num2)        | Returns the product of Num1 and Num2.                |
| Divide  | Num1.Divide(Num2)       | Returns the quotient; precision depends on settings. |
| Remdr   | Num1.Remdr(Num2)        | Returns the remainder of the division.               |
| Pow     | Num1.Pow(Num2)          | Calculates the power ( $x^y$ )                       |
| Sqrt    | Num1.Sqrt()             | Calculates the square root to current precision.     |
| Log     | Num1.Log()              | Calculate Logarithm to the base 10 ( $\log$ )        |
| Ln      | Num1.Ln()               | Calculates the Natural Logarithm ( $\ln$ ).          |
| Exp     | Num1.Exp()              | Calculates the Natural Exponent ( $e^x$ ).           |
| Fact    | Num1.Fact()             | Calculates the product of all positive integers.     |
| AbsVal  | Num1.AbsVal()           | Returns the absolute (non-negative) value.           |
| Neg     | Num1.Neg()              | Returns the value multiplied by -1.                  |
| Trunc   | Num1.Trunc()            | Returns the integer part without rounding.           |
| Fract   | Num1.Fract()            | Returns only the decimal/fractional part.            |
| Round   | Num1.Round(Scale)       | Rounds to a specified number of decimal places.      |
| Trim    | Num1.Trim()             | Trim leading and trailing 0                          |
| IsZero  | Num1.IsZero()           | Num1 is equal to zero (=0)                           |
| IsNeg   | Num1.IsNeg()            | Num1 is negative (<0)                                |
| IsPos   | Num1.IsPos()            | Num1 is positive (>0)                                |
| IsEq    | Num1.IsEq(Num2)         | Num1 is equal to (=) Num2                            |
| IsNEq   | Num1.IsNEq(Num2)        | Num1 is not equal to ( $\neq$ ) Num2                 |
| IsGT    | Num1.IsGT(Num2)         | Num1 is greater than (>) Num2                        |
| IsLT    | Num1.IsLT(Num2)         | Num1 is less than (<) Num2                           |
| IsGE    | Num1.IsGE(Num2)         | Num1 is greater than or equal to ( $\geq$ ) Num2     |
| IsLE    | Num1.IsLE(Num2)         | Num1 is less than or equal to ( $\leq$ ) Num2        |
| CmpTo   | Num1.CmpTo(Num2)        | Returns -1 (less), 0 (equal), or 1 (greater).        |

Table 6-1: BigDecimal Methods

## 6.2 BigDecimal Properties

The **BigDecimal** class includes a set of public properties that provide critical insights into the state and value of your numerical objects. Unlike standard numeric types in VBA, which require complex parsing to analyse, these properties allow you to directly access the internal characteristics of a number—such as its string representation, and its current scale (the number of digits after the decimal point). By utilizing these properties, developers can easily integrate high-precision results into the Excel UI or use them as conditional logic in complex financial and scientific algorithms.

| Property                  | Access     | Notes                                                                                                                                                                     |
|---------------------------|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>.Value</code>       | Read/Write | Gets or sets the BigDecimal object itself. This is the default property called when no property is specified.                                                             |
| <code>.ValueEx</code>     | Read Only  | Gets the value cast to a <code>vbInteger</code> , <code>vbLong</code> , <code>vbLongLong</code> , <code>vbSingle</code> , <code>vbDouble</code> , <code>vbString</code> . |
| <code>.TrScale</code>     | Read/Write | Gets or sets the number of decimal places for a BigDecimal number. Setting this can truncate or pad the number. Default is 100 decimal places.                            |
| <code>.AutoRound</code>   | Read/Write | A boolean flag that determines if the BigDecimal number should be automatically rounded. Default is <code>True</code> .                                                   |
| <code>.OutputDebug</code> | Read/Write | Enables internal logging to the VBA Immediate Window (Ctrl+G). Default is <code>False</code> .                                                                            |
| <code>.Pi</code>          | Read Only  | Returns the constant Pi as a high-precision BigDecimal object.                                                                                                            |

Table 6-2: BigDecimal Properties

### 6.2.1 The Default .Value Property

The `.Value` property is configured as the **Default Member** of the `BigDecimal` class. This means the property is automatically invoked when the object is referenced without an explicit property call. This design pattern allows for cleaner, more intuitive code that mimics the behaviour of native VBA variables.

' Instead of explicit property calls:

```
Num1.Value = Num2.Value
```

' You can simply write:

```
Num1 = Num2
```

In the example above, the default property mechanism handles the transfer of data: `Num2` yields its value, and `Num1` accepts it. This syntactic sugar is particularly powerful when passing `BigDecimal` objects into standard Excel functions or performing quick assignments within loops.

### 6.2.2 The .ValueEx Property

While the default .Value property is ideal for general assignments, the .ValueEx property provides a typed interface for extracting data. This property accepts a single parameter—a standard VBA VbVarType constant—to determine the format of the return value.

#### Syntax

```
result = BigDecimal.ValueEx(ReturnType)
```

#### Supported Return Types

| Constant                | VBA Type | Use Case                                                |
|-------------------------|----------|---------------------------------------------------------|
| <code>vbInteger</code>  | Integer  | Small whole numbers (-32,768 to 32,767).                |
| <code>vbLong</code>     | Long     | Standard 32-bit integer operations.                     |
| <code>vbLongLong</code> | LongLong | 64-bit integer operations (Available in 64-bit Office). |
| <code>vbSingle</code>   | Single   | Low-precision floating point.                           |
| <code>vbDouble</code>   | Double   | Standard Excel cell values and math functions.          |
| <code>vbString</code>   | String   | Maintaining full precision for display or storage.      |

Table 6-3: .ValueEx Property VBA Types

#### Example Usage

```
Dim MyStr As String
Dim MyDb1 As Double

' Retrieve as a string to preserve every decimal place
MyStr = Num1.ValueEx(vbString)

' Retrieve as a double for use in standard VBA math
MyDb1 = Num1.ValueEx(vbDouble)
```

### 6.2.3 The .TarScale Property

The .TarScale property determines the maximum number of digits allowed after the decimal point. This property is crucial for managing computational overhead and ensuring that operations like division—which can result in infinite repeating decimals—have a defined termination point.

Unlike standard VBA types that have a fixed precision, BigDecimal allows you to expand or contract this scale dynamically.

#### Usage and Behavior

- **Expansion:** Increasing .TarScale will effectively "pad" the number with trailing zeros, preparing it for higher-precision arithmetic.

- **Contraction:** Decreasing `.TarScale` will trigger a precision adjustment. Depending on the internal logic, the number will be truncated or rounded to fit the new scale.
- **Read/Write:** This property can be checked to verify current precision or set to enforce a specific scale before a calculation.

### Example: Controlling Precision

```
Dim Num1 As New BigDecimal
Num1 = "10.55555"

' Set the scale to 2 decimal places
Num1.TarScale = 2

Debug.Print Num1.ValueEx(vbString)      ' Output: 10.55 (truncation)

' Increase scale for scientific precision
Num1.TarScale = 10
Debug.Print Num1.ValueEx(vbString)      ' Output: 10.55
```

**Performance Tip:** While `BigDecimal` can handle very high scales, setting an unnecessarily large `.TarScale` (e.g., 1,000+) for simple calculations may impact performance. It is best practice to set the scale to the minimum required for your specific use case.

#### 6.2.4 Auto-Casting of Standard Types

To further enhance usability, the `BigDecimal` class supports **Auto-Casting** (Implicit Conversion). When a standard VBA numeric type or string is passed to a `BigDecimal` method (see Table 6-2) or assignment, the class automatically converts the input into a high-precision object before processing.

This eliminates the need to manually wrap literal values or standard variables, making arithmetic operations feel natural and concise.

Supported Types for Auto-Casting:

- **Integers:** `vbInteger`, `vbLong`, `vbLongLong`
- **Floating Point:** `vbSingle`, `vbDouble`
- **Strings:** `vbString` (Parsed directly into the internal format)

### Example: Simplified Arithmetic

```
' Without auto-casting, you would have to write:
' Num1 = Num1.Add(New_BigDecimal("1"))

' With auto-casting, standard types are handled instantly:
Num1 = Num1.Add(1)           ' Long is auto-cast
Num1 = Num1.Multiply(0.5)    ' Double is auto-cast
Num1 = "100.50"              ' String is auto-cast via the default .Value property
```

**Developer Note:** Auto-casting ensures that even when interacting with standard VBA types, the precision of the existing **BigDecimal** object remains the priority, preventing accidental "precision leakage" from the standard **Double** type until the operation is complete.

**Precision Warning:** When using `.ValueEx(vbDouble)` or `.ValueEx(vbSingle)`, values may be rounded or truncated to fit the limitations of the native VBA type. For values exceeding the range of a **Long** or **Double**, an overflow error may occur unless handled within your implementation.

#### 6.2.5 Effortless Precision with Auto-Trimming

The **BigDecimal** class is designed to deliver clean, professional output without the need for manual string manipulation. While the internal object may retain leading and trailing zeros to preserve the integrity of its data structure, the `.ValueEx(vbString)` property handles the cleanup for you. Every time you access the value via this property, the library automatically trims any non-significant zeros, ensuring your results are presented in their most concise and readable form while maintaining the high-precision accuracy you require.

#### 6.2.6 Mutable vs. Immutable Properties

Most properties in the **BigDecimal** class are **Mutable**, meaning you can change the value of an existing object without needing to create a new one using the factory function.

- **Direct Modification:** By setting `.StrValue` or `.DblValue`, you can reuse an existing **BigDecimal** variable and update its internal data instantly.
- **Precision Control:** Manually setting the `.Scale` allows you to force a specific decimal length on an existing number, which is useful for formatting financial data (e.g., forcing 2 decimal places).
- **The Pi Exception:** Unlike the other properties, **Pi** is a constant. It cannot be overwritten, as it represents a fundamental mathematical value. Attempting to set `Num1.Pi = 3` will result in a compilation error, ensuring the integrity of your geometric calculations.

### 6.3 Global Configuration with **BigOption**

The **BigOption** class provides a centralized mechanism for managing the behavior of all **BigDecimal** objects within your application. By setting properties on this global object, you define the default environment for any new **BigDecimal** instantiated thereafter.

This "seamless management" approach ensures that you don't have to manually configure every variable, reducing boilerplate code and preventing precision mismatches in large-scale projects.

## Supported Global Options

| Property         | Description                                                                                                                                                                            |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| .TarScale        | Sets the default number of decimal places for all newly created BigDecimal objects. Default is 100.                                                                                    |
| .AutoRound       | A boolean flag that determines if numbers should be automatically rounded (e.g., Round Half-Up) or truncated when they exceed the TarScale. Default is True.                           |
| AddTrailingZeros | A boolean flag that determines if a number is padded with zeros during a .Round operation. This flag requires the AutoRound flag to be True before a number will be padded with zeros. |
| .OutputDebug     | Enables internal logging to the VBA Immediate Window (Ctrl+G). Useful for tracing complex calculation steps or identifying performance bottlenecks. Default is False.                  |

Table 6-4: BigOption Properties

### 6.3.1 Active State Management

The **BigOption** class is not merely a collection of defaults; it is a **Live Manager**. Using an internal dictionary, it maintains a registry of every active BigDecimal instance within the project. This allows for dynamic, real-time updates across the entire application state.

#### Key Mechanisms:

- **Global Synchronization:** When you change a property on BigOption (such as .TarScale), the change is instantly propagated to every existing BigDecimal object currently in memory.
- **Lifecycle Awareness:** The manager intelligently tracks the lifecycle of your objects. When a BigDecimal is terminated or set to Nothing, BigOption removes it from the registry to prevent memory leaks and ensure that only valid, active objects receive updates.
- **Real-Time Control:** This architecture allows you to change the behaviour of your algorithms on the fly. For example, you can increase the precision of all active variables right before a sensitive calculation and lower it immediately afterward to optimize performance.

## Example: Real-Time Global Updates

```
Dim Num1 As New BigDecimal, Num2 As New BigDecimal
Num1 = 1.12345
Num2 = 9.87654

' Both Num1 and Num2 are currently at default TarScale (e.g., 20)
Debug.Print Num1.ValueEx(vbString) ' Outputs 1.123450...

' Update every active BigDecimal in the project instantly
BigOption.TarScale = 2

Debug.Print Num1.ValueEx(vbString) ' Now outputs 1.12
Debug.Print Num2.ValueEx(vbString) ' Now outputs 9.87
```

**Advanced Note:** Because BigOption manages the collection via a dictionary, the "Auto-Casting" and "Default Value" features remain lightning-fast. The manager ensures that the overhead of tracking objects does not interfere with the high-speed arithmetic processing of the individual classes.

### 6.4 Performance vs Not Responding with Yield

The BigDecimal **Yield** class provides an essential cooperative multitasking mechanism designed to preserve application stability during intensive, high-precision mathematical operations. Because arbitrary-precision arithmetic can require deep, multi-layered processing loops across the core BigDecimal and BigOption modules, execution can easily monopolize the **single-threaded** Excel interface, triggering a frustrating "Not Responding" status in the window caption. To counter this, the `.Events` method is called directly within the execution loops to increment the internal counter. Once this counter hits a predefined threshold, the method intercepts the cycle and fires a native VBA `DoEvents` instruction, allowing Excel to safely refresh its UI and process pending operating system messages.

Empirically, a default threshold of 500 iterations establishes an optimal performance equilibrium—frequent enough to keep the interface responsive, yet sparse enough to avoid the execution overhead that comes with over-polling the operating system. To ensure complete developer control, the class exposes a `.MaxCount` property allowing this 500-iteration benchmark to be dynamically tailored to the host machine's processing power, alongside a `.Clear` method to seamlessly reset the loop tracking metrics back to zero before starting a new calculation chain.

| Method               | Access     | Notes                                                |
|----------------------|------------|------------------------------------------------------|
| <code>.Events</code> | Read/Write | Used inside loops to increment the internal counter. |
| <code>.Clear</code>  | Read/Write | Called to clear the internal counter.                |

Table 6-5: Yield Class Methods



| Property               | Access     | Notes                                                                                               |
|------------------------|------------|-----------------------------------------------------------------------------------------------------|
| <code>.MaxCount</code> | Read/Write | Gets or sets the default loop iteration threshold before DoEvents is triggered. The default is 500. |

Table 6-6: Yield Class Property

## 7 The Automated Test Suite: Architecture and Execution

The automated test suite serves as the structural scaffolding for the BigDecimal library, acting as a rigorous, real-time quality control system. Implemented directly within the VBA environment, the suite operates by executing a sequential matrix of 69 distinct micro-assertions that isolate and stress every layer of the mathematical kernel.

It works by programmatically injecting highly volatile string inputs—such as extreme asymmetric scales, multiple leading or trailing zeros, explicit signs, and infinite repeating patterns—directly into the parser. The engine then processes these values through its underlying array-manipulation routines.

Upon completion of each operation, the test suite intercepts the internal high-precision array data, extracts the visible representation via a string casting method (`.ValueEx`), and performs a strict, character-by-character validation against verified, high-fidelity mathematical benchmarks.

By systematically running these checks across all primary operations (+, −, ×, ÷), sign states, logic gates, and complex transcendentals, the test harness instantly exposes any structural degradation, precision leaks, or broken internal dependencies across the entire code base.

### 7.1 What a 69/69 All-Green Pass Signifies

Achieving a flawless **69 / 69 "All Green"** pass rate shows the BigDecimal library is a robust, production-grade mathematical engine. In arbitrary-precision development, adding advanced layers like fractional exponentiation (`.Pow`) or long division (`.Divide`) frequently introduces subtle regression bugs that silently break core routines like addition and subtraction. Surviving this aggressive matrix shows that the entire library is perfectly integrated and structurally sound.

From an application standpoint, this means the engine is ready for deployment:

- **Financial Readiness:** Standard programming types suffer from the native IEEE 754 floating-point limitation, which truncates data at roughly 15 to 16 digits. In global enterprise environments or compound financial models, these tiny, hidden truncation errors roll up into massive balance discrepancies over millions of iterations. This

library ensures absolute transactional integrity by executing ripple-rounding carries and zero-annihilations.

- **Scientific Readiness:** For high-fidelity simulations, engineering models, or statistical computations, the engine maintains character-for-character fidelity with absolute mathematical reality out to dozens of arbitrary decimal scales.

Ultimately, this complete pass marks the library's evolution from an experimental programming project into a bulletproof, institutional-grade mathematical kernel.

## 7.2 How to run the Test Suite

The comprehensive test suite is integrated directly into the **BigDecimal Macro Source** spreadsheet. To validate the engine, simply execute the `RunAllTests` macro, which streams live verification outputs to the VBA **Immediate Window**. For a complete log of the pre-release deployment run, please refer to Appendix A – Test Suite Results for the official benchmark log recorded prior to production release.

## 8 Conclusion

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Thank you for choosing **BigDecimal**.

The goal of this library is to provide a robust, high-precision numerical engine that bridges the gap between standard VBA data types and the rigorous accuracy required for modern financial and scientific applications.

I hope the current feature set—from the intelligent **auto-trimming** logic to the definitive precision control of **TarScale**—empowers you to build more reliable and professional tools. As the development of this class continues, I am committed to maintaining this balance of mathematical integrity and developer-friendly flexibility.

## 9 Appendix A – Test Suite Results

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The following log represents the official baseline verification run of the BigDecimal mathematical kernel, executed in the VBA Immediate Window immediately prior to the production release.

=== STARTING AUTOMATED TESTS =====

```
[+] Basic Addition ..... [PASS]
[+] Addition Carry-over ..... [PASS]
[+] Massive Asymmetric Addition ..... [PASS]
[+] Cascading Ripple Borrow ..... [PASS]
[+] Sign Reversal Subtraction ..... [PASS]
[+] High-Precision Sign Reversal ..... [PASS]
[+] Double Negative Addition Route ..... [PASS]
[+] Subtracting From Negative ..... [PASS]
[+] Perfect Zero Annihilation ..... [PASS]
[+] Scale Alignment Multiplication ..... [PASS]
[+] Cascading Carry Multiplication ..... [PASS]
[+] Exact Integer Division ..... [PASS]
[+] Terminating Decimal Division ..... [PASS]
[+] Repeating Decimal (1/3) ..... [PASS]
[+] Repeating Decimal Loop (22/7) ..... [PASS]
[+] Small Divided by Large ..... [PASS]
[+] Positive Divided By Negative ..... [PASS]
[+] Negative Divided By Negative ..... [PASS]
[+] Division Edge Round-Up (2/3) ..... [PASS]
[+] Log10(100) = 2 ..... [PASS]
[+] High-Precision Ln(2) ..... [PASS]
[+] Exp(0) = 1 ..... [PASS]
[+] Identity Exp(Ln(5)) = 5 ..... [PASS]
[+] High-Precision Exp(1) [e] ..... [PASS]
[+] IsZero on True Zero ..... [PASS]
[+] IsPos on True Zero ..... [PASS]
[+] IsNeg on True Zero ..... [PASS]
[+] IsZero on Tiny Pos ..... [PASS]
[+] IsPos on Tiny Pos ..... [PASS]
[+] IsNeg on Tiny Pos ..... [PASS]
[+] IsZero on Neg ..... [PASS]
[+] IsPos on Neg ..... [PASS]
[+] IsNeg on Neg ..... [PASS]
[+] IsEq (Trailing Zeros Ignored) ..... [PASS]
[+] IsNEq (Trailing Zeros Ignored) ..... [PASS]
[+] IsEq on Mismatch ..... [PASS]
[+] IsNEq on Mismatch ..... [PASS]
[+] IsLT (Strictly Less) ..... [PASS]
[+] IsGT (Strictly Greater) ..... [PASS]
[+] IsLE (Distinct values) ..... [PASS]
[+] IsGE (Distinct values) ..... [PASS]
[+] IsLT when Equal ..... [PASS]
[+] IsGT when Equal ..... [PASS]
[+] IsLE when Equal ..... [PASS]
[+] IsGE when Equal ..... [PASS]
[+] Log10 Negative Guard Rail ..... [PASS]
[+] Rounding UP without leading digits before dot ..... [PASS]
[+] Rounding UP Test ..... [PASS]
[+] Rounding DOWN Test ..... [PASS]
[+] Embedded Internal Zeros ..... [PASS]
[+] Division by Zero Guard Rail ..... [PASS]
[+] Ln(0) Guard Rail ..... [PASS]
[+] Ln(Negative) Guard Rail ..... [PASS]
[+] AbsVal Negative Input ..... [PASS]
[+] Neg Positive Input ..... [PASS]
[+] Neg Negative Input ..... [PASS]
[+] Truncate Pulls Integer ..... [PASS]
```

[+] Frac Pulls Fraction Part ..... [PASS]  
[+] Truncate Negative ..... [PASS]  
[+] Frac Negative ..... [PASS]  
[+] Parser Leading/Trailing Zeros ..... [PASS]  
[+] Parser Explicit Plus Sign ..... [PASS]  
[+] Round to 4 places ..... [PASS]  
[+] Remainder check ..... [PASS]  
[+] Factorial of 5 ..... [PASS]  
[+] Factorial Large Stress (30!) ..... [PASS]  
[+] High-Precision Sqrt(2) ..... [PASS]  
[+] Pow Integer Base/Exp ..... [PASS]  
[+] Pow Fractional Base/Exp ..... [PASS]

=====

TESTS COMPLETE: 69 / 69 PASSED.  
\*\*\* ALL GREEN! No features broken.